

EMAIL THREAT FORENSIC INVESTIGATION REPORT

OFFICIAL DIGITAL FORENSIC EXAMINATION DOSSIER // LAW ENFORCEMENT & SOC RECORD

Case Identifier	Examination Date	Investigation Status	Evidence Hash (SHA-256)
CASE-005	2026-09-05	VERIFIED EXAMINATION	1f8e5b2a9c4d7063e8a1f5c2d9b6...
Classification: PAYMENT_D...	Risk Score: 81 / 100	Sample: Updated_Bank_Deta...	File Size: 94 KB

1.0 EXECUTIVE SUMMARY

A forensic examination was conducted on evidence sample 'Updated_Bank_Details_Invoice_88421.eml' cataloged under reference CASE-005. The electronic message claims transmission from sender 'Suresh Iyer' (accounts@apex-logistics.co.in). Automated and deterministic protocol evaluation assessed an authoritative risk score of 81 out of 100, resulting in a forensic threat classification of PAYMENT_DIVERSION. Authentication protocol verification recorded SPF: SOFTFAIL, DKIM: FAIL, and DMARC: FAIL. The message was accompanied by an identified lookalike typosquat target domain (apexlogistics.co.in), indicating intentional brand impersonation. Routing telemetry analyzed 3 relay hops from origin to terminal destination server.

2.0 EMAIL & EVIDENCE SPECIFICATIONS

Evidence Field	Forensic Data Value	Verification Finding
Original Filename	Updated_Bank_Details_Invoice_88421.eml	Binary structure parsed and pres...
File Size	94 KB	Standard envelope size
Cryptographic Hash	1f8e5b2a9c4d7063e8a1f5c2d9b6e0a4c7f3d1b8e5a2c9f...	SHA-256 cryptographic integrity ...
Claimed Sender Name	Suresh Iyer	Header-extracted display name
Claimed Sender Email	accounts@apex-logistics.co.in	Header From address
Terminal Mail Server	Pune, India	Terminal destination host
Originating Node	89.32.148.77	First recorded external IP hop

3.0 AUTHENTICATION PROTOCOL ANALYSIS

Standard email authentication protocols were checked against published Domain Name System (DNS) records. Discrepancies indicate that the sending infrastructure was not authorized to transmit on behalf of the domain.

Protocol	Verdict	Technical Examination Finding
SPF (Sender Policy Framework)	SOFTFAIL	Softfail / record neutral
DKIM (DomainKeys Identified)	FAIL	Cryptographic signature check: Missing or invalid public key cryp...
DMARC (Domain Alignment)	FAIL	Domain alignment policy check: Failed alignment criteria with pub...

4.0 HEADER & ROUTING INFRASTRUCTURE

The message envelope traversed 3 recorded network transit hops. Network infrastructure was cross-referenced with autonomous system records and datacenter threat feeds.

Hop	IP Address	Geographic Location	Relay Timestamp	Forensic Infrastructure Note
#1	89.32.148.77	Bucharest, Romania	2026-09-06 07:30:15	Bulletproof hosting origin
#2	103.86.99.21	Singapore	2026-09-06 07:30:41	Transit MX
#3	103.25.60.8	Pune, India	2026-09-06 07:31:02	Destination mail server

5.0 LANGUAGE, INTENT & CONTENT ANALYSIS

Content linguistics and social engineering vectors were scored at 28 out of 30 maximum allocated points. Extracted textual phrases were analyzed for deception patterns, urgency cues, and financial solicitation.

Category / Signal	Detected Phrasing / Pattern	Confidence Assessment
FINANCIAL DIVERSION	"remit all outstanding invoices to our updated bank account"	92.0% Match
DECEPTION	"do not use previous banking instructions due to ongoing a...	86.0% Match

6.0 REPUTATION & DOMAIN INTELLIGENCE

Attribute	Observed Evidence	Analyst Interpretation
Domain Age	21 days	High anomaly: Domain registered immediately pr...
Typosquatting Target	apexlogistics.co.in	Deliberate brand impersonation targeting legit...
Reputation Score Contribution	11 / 15 pts	Elevated risk: Associated infrastructure flagg...

7.0 INDICATORS OF COMPROMISE (IOCs)

The following actionable technical indicators were extracted during examination for perimeter blocklisting and SIEM correlation:

Indicator Type	Technical Value	Context / Threat Finding
IP Address (Origin)	89.32.148.77	Originating relay host observed in he...
Email Address	accounts@apex-logistics.co.in	Header From address
Typosquat Target	apexlogistics.co.in	Target brand impersonated in attack
Forensic Artifact	IFSC:YESB0000441	Extracted payment/URL artifact
Forensic Artifact	Reply-To:payments@apex-logistics-billing.com	Extracted payment/URL artifact
File Hash (SHA-256)	1f8e5b2a9c4d7063e8a1f5c2d9b6e0a4c7f3d1b8e5a2c...	Unique cryptographic hash of message ...

8.0 ATTRIBUTION ASSESSMENT

ATTRIBUTION FINDING: ANONYMIZED / HOSTING INFRASTRUCTURE

Origin infrastructure utilized bulletproof or datacenter hosting to obscure attacker physical identity.
Note: Attribution conclusions represent technical forensic assessments based on available telemetry and do not constitute definitive legal identification without formal judicial subpoena records.

9.0 CORRELATION & INFRASTRUCTURE RELATIONSHIPS

Graph analysis cross-referenced entities across historical telemetry clusters and previous incident dossiers:

Source Entity	Target Entity	Correlation Basis / Linkage
email1	ip1	same sending IP
email1	domain1	mismatched reply-to
email2	email1	invoice thread clone

10.0 RISK ASSESSMENT & SIGNAL WEIGHTING

Overall Risk Score: 81 / 100 (Threshold evaluation: CRITICAL RISK). The score is derived across four standardized examination quadrants:

Quadrant Vector	Allocated Weight	Observed Score	Analytical Contribution
Header & Routing	30 pts	20 pts	Evaluates relay anomaly, ASN hosting, and h...
Authentication Failure	25 pts	22 pts	Evaluates SPF, DKIM signature validity, and...
Language & Intent	30 pts	28 pts	Evaluates urgency, financial diversion, and...
Reputation & History	15 pts	11 pts	Evaluates domain age, typosquatting targets...

11.0 KEY INVESTIGATION FINDINGS

- High-confidence threat signature: Overall threat score exceeds enterprise containment threshold.
- Message failed authentication verification: Sending host lacks authoritative domain alignment.
- Lookalike brand impersonation identified: Attacker targeted legitimate entity 'apexlogistics.co.in'.
- Origin routing: Outbound transmission originated from standard commercial/workspace relay.

12.0 RECOMMENDED INVESTIGATION ACTIONS

- Evidence Preservation: Retain full raw RFC 5322 message headers and message body in forensic storage.
- Network Containment: Implement perimeter firewall block on originating IP 89.32.148.77.
- DNS Mitigation: Issue internal DNS sinkhole rule for identified typosquatting target apexlogistics.co.in.
- Incident Response: Query mail gateway logs for all inbound messages matching the sending domain within past 30 days.
- Account Safeguards: Enforce credential reset and session termination for any recipient who engaged with message content.

13.0 AUTOMATED NARRATIVE GENERATION ARCHITECTURE NOTE

SYSTEM NOTICE: This document was produced via deterministic evidence extraction from validated email forensic JSON. The platform architecture is designed to integrate text-generation language models in future backend iterations to draft extended contextual DFIR narratives while maintaining the underlying JSON as the single source of truth.

14.0 EVIDENCE ATTESTATION & LEGAL DISCLAIMER

REPORT IDENTIFIER: REP-CASE-005 | **PLATFORM:** Email Threat Intelligence & Forensic Platform | **EVIDENCE SOURCE:** Ingested MIME/MSG Artifact | **STATUS:** VERIFIED FORENSIC RECORD

LEGAL DISCLAIMER: This document is an analytical examination record compiled from electronic message headers, cryptographic authentication tokens, DNS telemetry, and network routing hops. Findings should be corroborated with original raw headers, provider server logs, and judicial legal process before presentation in legal proceedings.